

Education

Monash University

*Department of Econometrics and
Business Statistics*

Doctor of Philosophy

Jan 2023 - present

- My research interests falls in financial econometrics, Bayesian analysis and asset pricing. My current research project bridge the Bayesian non-parametric analysis to the empirical finance data, with particular focus on the price jumps.

University of Melbourne

Melbourne Business School

Doctor of Philosophy

June 2022 - Jan 2023

- I started my PhD at Melbourne Business School and then transferred to Monash University to continue my PhD degree.

Monash University

Monash Business School

Bachelor of Commerce (1st Class Honors)

Feb 2017 - Dec 2020

- Major in Finance and Econometrics
- Graduated with High Distinction
- GPA: 3.9/4.0

Research Experience

Jump Modeling with Dirichlet Process Mixture

Monash University & University of Melbourne

June 2022 - Present

- **Abstract:** We propose a Bayesian nonparametric method using Dirichlet process mixture to model jumps. This new model encompasses existing parametric jump specification, but allows for extreme jumps to occur in either tail of the return distribution via infinite mixture model. The flexibility in our model specification allows for multi-modal jump size distribution. We investigate the robustness of our proposed model under different jump intensity and size specifications under simulations and provide comparisons about the model-driven option implied volatility smiles. We verify the practical use of our model with empirical data in jump detection, data-driven jump size density and implied volatility smiles.

Optimal probabilistic forecasts for risk management

Monash University

Mar 2020 - Mar 2023

- **Abstract:** This paper explores the implications of producing forecast distributions that are optimized according to scoring rules that are relevant to financial risk management. We assess the predictive performance of optimal forecasts from potentially misspecified models for i) value-at-risk and expected shortfall predictions; and ii) prediction of the VIX volatility index for use in hedging strategies involving VIX futures. Our empirical results show that calibrating the predictive distribution using a score that rewards the accurate prediction of extreme returns improves the VaR and ES predictions. Tail-focused predictive distributions are also shown to yield better outcomes in hedging strategies using VIX futures.
- **Link to the paper:** <https://arxiv.org/abs/2303.01651>

Talks

Econometric Society Australian Meetings (ESAM)

Monash University, Australia

December, 2024

Time Series and Forecasting Symposium

University of Sydney, Australia

November, 2024

- Poster presentation
- Best Poster Runner-up

Work Experience

Monash Business School

Melbourne, Australia

Teaching Associate

Feb 2023 - present

- High Dimensional Data Analysis (Head tutor)
- Data Analysis in Business (Excellent teaching team)
- Introductory Econometrics
- Business Statistics
- Business Forecasting

Monash Business School

Melbourne, Australia

Research Assistant

Aug 2023 - Oct 2024

- High frequency data analysis of bond data for the Thailand Bond Market Association. Evaluate bond price jumps under the influence of regulation changes.

Melbourne Business School

Melbourne, Australia

Research Assistant

Aug 2022 - Jan 2022

- Use Python to extract high-frequency data of each stock in S%P 500 list from database - Refinitiv
- Using R to finish the data cleaning job, e.g. Extract the five-minute price for every stock and aggregate high frequency return to daily level.

Honors & Awards

2024	Best Poster Runner Up , Time Series and Forecasting Symposium, University of Sydney
2023	Co-funded Monash Graduate Scholarship , Monash University
2022	Faculty of Business and Economics Graduate Research Scholarship , University of Melbourne
2021	2020 Dean’s Honor (First Class) , Monash Business School
2020	Econometrics Memorial Scholarship , Department of Econometrics and Business Statistics, Monash
2020	Meritorious Winner , 2020 Interdisciplinary contest in Modelling